# CODE HILLS 1001 — Business Plan

### Kirinda Grain Guard — Umuravumba

> Part of the CODE HILLS 1001 business-plan library (6 plans total, built from Rwanda's natural resources: Umuravumba, Tea, Coffee, Umwenya, Sisal). Figures in RWF are labeled estimates — verify current local prices, supply arrangements, and regulatory requirements (RDB/RFDA/technical institute) before investing.

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## LOW-CAPITAL BUSINESS 1 — "Kirinda Grain Guard" (Umuravumba)

### 1. Business Identity

- **Name:** Kirinda Grain Guard
- **Tagline:** *"What protected our grandmother's grain now protects your harvest."*
- **Resource:** Umuravumba (*Tetradenia riparia*), the strong-smelling shrub traditionally planted around Rwandan homes.
- **Concept in one sentence:** Dried, bundled, and sachet-packed umuravumba leaf products that protect stored grain, beans, and household spaces from insects and mosquitoes, sold to farmers, cooperatives, and households as a low-cost, chemical-free alternative to synthetic pesticide dust.
- **What makes it unique:** It is not a perfume or a soap. It is positioned and sold as a **post-harvest loss-prevention input** through agro-dealers and cooperatives (an agricultural sales channel), with a household mosquito-comfort sachet as a secondary product — a dual-market structure almost no "umuravumba perfume" idea would arrive at.

### 2. Problem

Two real, well-documented Rwandan problems, solved by one raw material: - **Post-harvest loss:** Smallholder farmers regularly lose stored beans, maize, and sorghum to storage pests (weevils, moths) between harvest and sale/consumption, especially where hermetic (airtight) storage bags are not used or affordable. **[ASSUMPTION — VERIFY LOCALLY: exact loss percentages vary by district and crop; the entrepreneur should ask a local cooperative or Rwanda Agriculture Board (RAB) extension agent for current figures before pricing a solution.]** - **Household mosquito discomfort:** Many households, especially in areas without reliable window screening, want an affordable, chemical-free way to reduce mosquito presence in living spaces, particularly in the evening.

### 3. Solution

Dried umuravumba leaf is bundled and layered into grain storage bags/granaries (a traditional practice already known informally in some communities) in a **standardized, tested, pre-dried, pest-ready format** — something households currently do inconsistently with fresh leaves cut from their own hedge. Kirinda Grain Guard turns an informal home practice into a **reliable, packaged, quality-controlled product** with clear usage instructions, sold alongside a household sachet line for living spaces.

### 4. Product

- **Grain Guard Bundles:** Dried, coarsely crushed umuravumba leaf packed in breathable woven-sisal or paper mesh pouches (150g), designed to be layered inside grain sacks or granaries. Sold in multi-packs timed to harvest season.
- **Home Comfort Sachets:** Smaller dried-leaf sachets (30g) in breathable cotton/sisal-blend bags for wardrobes, under beds, and sitting areas — marketed for mosquito comfort and pleasant scent, not as a medical repellent (no health claims are made, to stay within safe, honest marketing).
- **Variations:** A "cooperative bulk" format (2kg loose dried leaf for larger granaries) sold directly to cooperative storage facilities.
- **Packaging/service format:** Breathable natural-fiber pouches only (no plastic film, which would trap moisture and cause mold — a real technical reason, not just an eco-slogan).
- **Why customers want it:** Free/cheap raw material already trusted informally in their own communities; clearly different from synthetic pesticide dust (no chemical smell, no dosage risk to handle).
- **Why it's hard to confuse with a generic product:** The core offer is *loss-prevention for stored grain*, sold through **agro-dealers and cooperatives**, not a fragrance retailer — a completely different shelf and a completely different buyer conversation than "essential oil" or "natural perfume" ideas.

### 5. Sustainability

- **Sourcing:** Umuravumba is a coppicing shrub — cutting branches for leaf harvest does not kill the plant and stimulates new growth, so no land clearing or replanting is required.
- **Waste minimization:** Stems left after leaf-stripping are dried and sold as kindling/cooking-fire starter to local eateries — a genuine byproduct reuse, not a marketing add-on.
- **Packaging:** Woven-sisal or plain paper mesh pouches, both biodegradable and, notably, made from **another of the five resources** (sisal), creating a natural cross-business input relationship if the entrepreneur also engages with Low-Capital Business 3.
- **Energy/water:** Sun-drying only (no fuel-based dryers needed in Rwanda's climate for most of the year); no water used in processing.
- **End-of-life:** Spent sachets/bundles are fully compostable and can be added directly to a household compost pile or granary floor mulch.
- **Path to zero-waste:** At full maturity, 100% of harvested plant material is used (leaf → product, stem → kindling, packaging →

compostable), leaving no processing waste stream.

### 6. Target Customers

- Smallholder farmers and farming households needing low-cost storage protection.
- Agricultural cooperatives with shared granaries/warehouses.
- Households seeking a natural mosquito-comfort product (urban and rural).
- Agro-dealers/input shops as a stocked product line.

### 7. Competitive Advantage

- **Advantage source:** A specialized customer problem (post-harvest loss) combined with a distribution channel (agro-dealers/cooperatives) that fragrance or wellness competitors would not naturally use.
- **Why competitors may not immediately copy this:** Most AI-assisted “umuravumba” ideas default to perfume, soap, or incense, sold through gift/retail channels. Copying this specific model requires building a **relationship with agro-dealers and cooperative storage managers** — a slower, less obvious, and less “exciting” channel to identify than a retail shelf, which discourages quick copycats.

### 8. Startup Requirements

**Essential:** - Access to umuravumba shrubs (own compound, neighbors, or purchase small bundles from those who have hedges) — near-zero cost. - Drying space (raised racks or mats under shade) — can be built from scrap wood. - Simple scale for weighing. - Basic sisal-mesh or cotton pouches (bought from local tailors/market or self-sewn). - Hand tools: machete/pruning shears, mortar or crushing board. - Local government business registration (RDB one-stop, low-cost for a sole proprietorship). **Add later:** - Branded printed labels. - A small moisture-proof storage cabinet for finished stock. - A motorcycle/bicycle for cooperative delivery routes.

### 9. Startup Budget (RWF) — all figures are estimates; verify current local prices

| Item | Minimum | Recommended |
|---|---:|---:|
| Pouches/mesh bags (200 units) | 40,000 | 80,000 |
| Drying rack materials | 0 (reused wood) | 25,000 |
| Scale | 8,000 | 15,000 |
| Hand tools | 5,000 | 12,000 |
| Business registration | 0–10,000 | 15,000 |
| Labels/branding | 0 (handwritten) | 20,000 |
| Initial transport (market visits) | 5,000 | 15,000 |
| **Total** | **≈58,000–68,000** | **≈182,000** |

### 10. Unit Economics (illustrative — verify locally)

- **Home Comfort Sachet (30g):** Cost to make ≈ 250 RWF (pouch + labor share) → Sell at 800 RWF → Gross profit ≈ 550 RWF (~69% margin).
- **Grain Guard Bundle (150g):** Cost ≈ 500 RWF → Sell at 1,500 RWF → Gross profit ≈ 1,000 RWF (~67% margin).
- At **10 units/day** (mixed): ≈7,500 RWF/day gross profit.
- At **30 units/day**: ≈22,500 RWF/day gross profit.
- At **100 units/day** (requires cooperative bulk orders, not just retail): ≈75,000 RWF/day gross profit.
- **Note:** these are gross figures before transport, packaging waste/spoilage, and the entrepreneur’s own time; net profit will be meaningfully lower, especially at low volume.

### 11. Production Process

1. Harvest branches (cut, don’t uproot).
2. Strip leaves; set stems aside for kindling.
3. Sun-dry leaves on raised, covered racks 3–5 days until fully dry and brittle (test: leaf crumbles, no bend).
4. Coarsely crush by hand or mortar (avoid powder, which can be dusty/irritating).
5. Weigh into pouches, seal, label with harvest date.
6. Store in a dry, ventilated space; discard any batch showing mold before packing. **Quality/safety notes:** No skin-contact or ingestion claims; keep away from small children if crushed fine; store away from moisture.

### 12. Marketing Strategy (first 30 days)

- **Week 1:** Visit 3 local agro-dealers and 2 cooperatives with free samples and a one-page spec sheet; post short WhatsApp/TikTok videos showing the traditional practice being “professionalized.”
- **Week 2:** Set up a stall at a local market day; collect phone numbers via a simple WhatsApp broadcast list.
- **Week 3:** Approach one cooperative for a small pilot order (e.g., 20 bundles) at a discount in exchange for a testimonial.
- **Week 4:** Use the testimonial in an Instagram/TikTok post; begin door-to-door household sachet sales in the entrepreneur’s own

neighborhood.

### 13. Sales Strategy

- **Where:** Agro-dealers (B2B), cooperatives (B2B), local markets and door-to-door (B2C).
- **Pricing:** Anchor slightly below synthetic alternatives where those exist, and above “free home practice” by emphasizing convenience/reliability.
- **First 10 customers:** Neighbors and cooperative contacts, offered a discount for feedback.
- **Feedback:** A simple paper or WhatsApp form asking “did you see fewer pests in your grain after 30 days?”
- **Model:** One-time purchase initially; move cooperative accounts to a **repeat seasonal contract** (pre-harvest restock) once trust is established.

### 14. Growth Plan

1. **Home production** — one household, hand tools.
2. **Small workshop** — hired part-time labor for harvesting/drying, dedicated drying shed.
3. **Local brand** — consistent packaging, RDB-registered brand, presence in multiple markets.
4. **Rwanda-wide distribution** — agreements with agro-dealer chains/cooperative unions across provinces.
5. **Regional opportunity** — supply to similar agro-ecological zones in neighboring EAC countries via existing cooperative-union trade links (a longer-term, unverified opportunity — **[ASSUMPTION]**).

### 15. Risks

| Risk | Mitigation |
|---|---|
| Leaf supply seasonality/availability | Cultivate own hedge rows; source from multiple households, not one supplier |
| Mold/quality failure in drying | Strict moisture testing before packing; covered racks |
| Low customer awareness of product category | Free trial samples; visible demonstrations |
| Competition from synthetic pesticide dust | Price and message on chemical-free safety, not head-to-head efficacy claims |
| Regulatory ambiguity (crop-protection product) | Avoid pesticide/medical claims; position as traditional storage aid; consult RAB/RDB before scaling |
| Cash flow (seasonal cooperative orders) | Keep household sachet sales running year-round as steady income |

### 16. 90-Day Launch Plan

- **Weeks 1–2:** Validate with 10 households/1 cooperative contact; confirm leaf access.
- **Weeks 3–4:** Build drying rack; produce first 100-unit test batch.
- **Weeks 5–6:** Test packaging formats; get feedback from 20 trial users.
- **Weeks 7–8:** Register business; finalize branding/labels.
- **Weeks 9–10:** First paid sales push (market + agro-dealer visits).
- **Weeks 11–12:** Secure first cooperative pilot order; set up simple record-keeping (income/expense notebook or spreadsheet); reinvest profit into a second drying rack.

### Originality Test

1. **Obvious idea a competitor might propose:** Umuravumba essential oil, perfume, soap, or incense sold at gift shops.
2. **How this is structurally different:** It targets a farming/storage problem, not a fragrance want, and sells through agro-dealers/cooperatives, not retail gift channels.
3. **Distinctive combination:** Resource (umuravumba) + problem (post-harvest loss) + customer (farmers/cooperatives) + channel (agro-input distribution) is a combination no “perfume” pitch would land on.
4. **Hard to copy because:** Requires real relationships with agro-dealers/cooperative storage managers, plus credible framing as a storage aid rather than a cosmetic — slower and less obvious to replicate than sourcing leaves and bottling oil.